

Decision Support System for Chemo

Maximum effect on the tumour, Minimum harm to the patient.

PROBLEM

Chemotherapy has to shrink the tumour without harming healthy organs, but the dose is usually lowered only **after** the patient already has side effects.

SOLUTION

Before a dose is given, the clinician sees what it is expected to do to the tumour and to **each major organ**, and which organ is limiting it.

IMPACT

The full course is reviewed and alternatives compared **before treatment starts**, so the dose is decided in advance rather than in reaction.

WHAT THE NUMBERS MEAN

Tumour cells: how much cancer is left, counted on a logarithmic scale.

Side-effect load: the model's own 0–100 scale of drug burden, not a clinical grade.

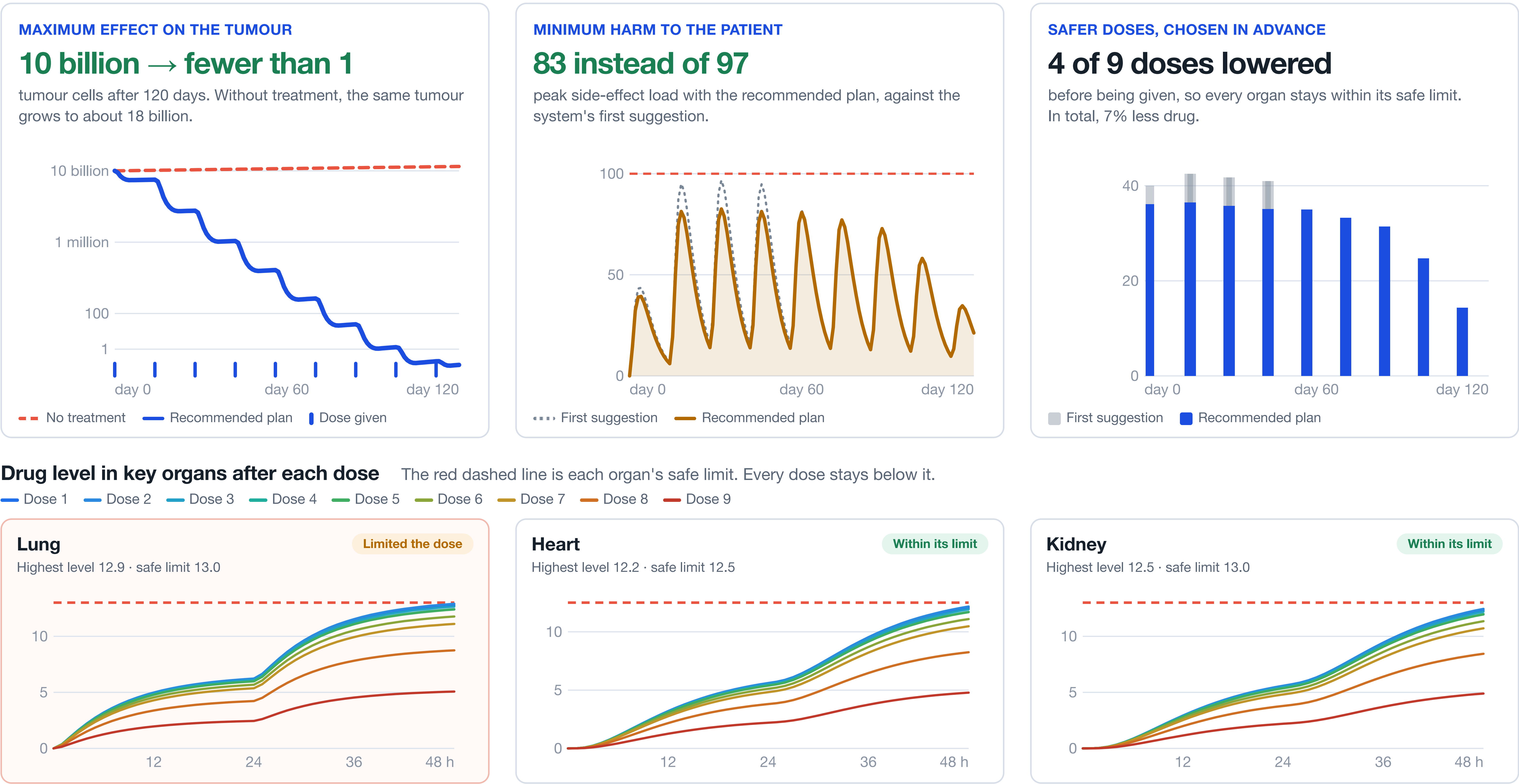
Safe limit: the most drug an organ should see in the 48 h after a dose.

Dose: the amount given on the cycle day and the day after, in the model's units.

A CASE STUDY

A 70 kg adult with bulky disease, treated for 120 days

About 10 billion tumour cells at the start (~10 g of tumour), a dose every 14 days, and standard adult organ limits. Every figure here is the system's own output for this case.



Why it matters for doctors

- Fewer calculation mistakes**
Body size and every dose adjustment are worked out automatically, not by hand.
- See the impact first**
The tumour's response and the side effects are projected over the whole course.
- 13 organs checked**
A dose that would push any organ past its safe limit is lowered before it is given.
- Thinks the way a doctor does**
Mimics how an oncologist reasons, and explains every dose in plain words.
- Personalised to each patient**
Adjusts for body size, tumour burden and organ tolerance.
- Compare before deciding**
Put schedules side by side and choose the best balance for the patient.

How it works: every step is finished before the patient is given anything

